
False Negative Control for Automated Tumour Segmentation

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Abstract

Deep learning has significantly improved tumour segmentation in medical imaging, yet the deployment of black-box models in clinical practice is limited by uncontrolled diagnostic risks. While accuracy is of importance, clinical decision-making demands certainty to confidently confirm or rule out harmful diagnoses. This paper proposes a framework to control the expected false negative rate of tumour segmentation systems with formal statistical guarantees through *Conformal Risk Control* and *Adaptive Conformal Prediction*. The net result is a method that offers improved interpretability and greater transparency, facilitating more informed clinical decision-making. Experiments on the BraTS2020 dataset, consisting of 368 MRI scans demonstrate that this approach successfully controls the false negative rate under 5% for most cases, while highlighting the reliance on the baseline segmentation confidence. Ultimately, this work showcases the potential of conformal prediction to enhance trust and reliability in the application of black-box models in medical imaging.

Introduction

Medical imaging plays a central role in modern diagnostics, with deep learning models now capable of achieving remarkable segmentation accuracy for tasks such as tumour detection. However, despite these advancements, the clinical adoption of such models remains hindered by their inability to provide reliable uncertainty estimates and formal guarantees on diagnostic risk. In high-stakes applications like tumour segmentation, the primary concern is not just accuracy but the assurance that no critical tumour regions go undetected. False negatives—where tumour tissue remains undetected and is misclassified as non-tumour—pose a significant risk, as they can lead to missed diagnoses and suboptimal treatment planning. Addressing this issue requires a framework that not only enhances segmentation performance but also provides formal guarantees on the false negative rate and ensures that the likelihood of missing tumour regions remains within a predefined threshold.

A solution to these concerns can be found in conformal prediction, a statistical framework that provides post hoc uncertainty quantification with finite-sample guarantees under relatively mild assumptions [8]. Conformal prediction has historically been studied in simpler classification and regression contexts, where it offers confidence sets (or prediction intervals) that include the true label with user-prescribed probability. Recently, researchers have shown how to adapt these methods to more complex tasks such as multi-class classification, semantic segmentation, and object detection.

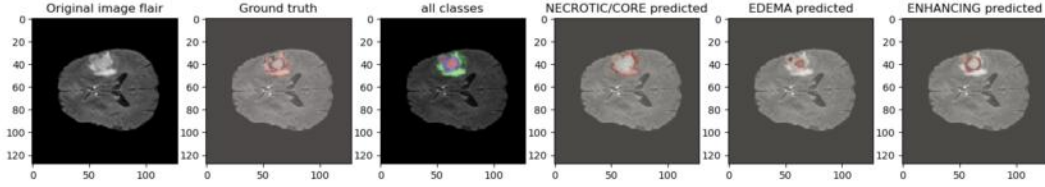


Figure 1: Basic tumour segmentation and classification

This work utilises the theoretical foundations of conformal prediction to produce segmentation masks for tumour detection and classification through Conformal Risk Control. Rather than focusing simply on coverage guarantees (i.e., ensuring that the true label is included in the predicted set at a given frequency), This work focuses on controlling the false negative rate (FNR). This enables the creation of models that can provide formal statements of the type “the expected fraction of undetected tumour voxels missed by the uncertainty masks is no more than α ,” where α is a user-selected threshold, such as 5%. This approach is of particular interest in medical imaging, where false negatives represent a critical failure mode, as these mistakes can lead to entirely overlooked regions in the tumour.

We base our practical demonstration on an existing black-box model developed by Rastislav [5], which uses a deep convolutional neural network for tumour segmentation. The conformal calibration method operates as a wrapper for these baseline predictions, extending them into segmentation masks that also include uncertainty sets. One might think of these uncertainty sets as “safety margins” around the initially segmented tumour region, informed by the model’s internal confidence scores and a calibration procedure that aligns the expected FNR with a target bound. Through this mechanism, when the baseline model is extremely confident about a particular region being tumour or non-tumour, the conformal sets remain small. Conversely, for ambiguous regions, the conformal sets expand, indicating uncertainty and prompting clinicians to pay additional attention to those areas.

The contributions of this paper are threefold. First, we provide a rigorous overview of how to translate the Conformal Risk Control mechanism to the tumour segmentation problem, ensuring that an external user-defined risk metric—here, the FNR—can be bounded in expectation by a prescribed α value. Second, we showcase how this approach can generate easily interpretable segmentation masks with expanded “uncertainty regions” that highlight potential areas of oversight or misclassification. Finally, we present a comprehensive empirical evaluation on a dataset of 368 magnetic resonance imaging (MRI) scans, illustrating the practical performance and limitations of our framework. The results demonstrate that, while we can maintain an average FNR below 5%, this comes at the cost of a modest increase in false positives and the reliance on the underlying model’s confidence estimates.

The paper proceeds as follows. We begin by reviewing related work on conformal prediction and discussing the theoretical underpinnings of Conformal Risk Control and Adaptive Conformal Prediction. We then present the details of our methodology, including how we define and calibrate our loss function (FNR) and the procedure for generating uncertainty segments around the baseline segmentation. Subsequently, we describe the dataset, model architecture, and calibration strategy, followed by the quantitative and qualitative results of our experiments. We then interpret these findings in a broader medical context, discuss the limitations of our approach, and suggest future directions for improvement. Finally, we conclude by reflecting on the significance of conformal methods for medical image segmentation and the potential for further integration of advanced uncertainty quantification strategies into high-risk domains.

1 Methods

Conformal Prediction

Conformal prediction [7, 8] provides a framework for constructing predictive sets with finite-sample guarantees. The fundamental premise is that, for a collection of i.i.d. samples, one can use a portion of the data (the calibration set) to determine a threshold such that the resulting set of predictions includes the true label at a rate matching a user-specified confidence level. The methodological novelty that forms the basis of our approach is Conformal Risk Control [1], a theoretical framework that allows control over both coverage and any monotonic loss function in expectation. At the same time, we

draw upon the notion of Adaptive Conformal Prediction [6], which permits more adaptively-sized conformal sets, effectively allowing each prediction to have its own calibrated threshold conditioned on local confidence scores.

In standard conformal prediction for multi-class classification, the model outputs a softmax probability distribution over the possible classes, and one calibrates a threshold on these probabilities to form a subset of classes that meet a coverage criterion. For tumour segmentation, the task is more granular and involves labelling every pixel (in 2D) or voxel (in 3D) of an image volume as tumour or non-tumour, and potentially subdividing tumours into different subtypes. The use of softmax outputs at the voxel level enables the direct extension of conformal methods, but the dimensionality of the problem introduces new complexities. When controlling a specific risk measure like FNR, one must ensure that, collectively across *all* voxels, the expected fraction of misclassified tumour voxels does not exceed α .

Formally, let $(X_1, Y_1), \dots, (X_n, Y_n)$ be i.i.d. samples, where X_i represents a $d \times d \times d$ MRI scan and the corresponding ground-truth segmentation and tumour classification mask $Y_i \in \{1, \dots, K\}^{d \times d \times d}$. A portion of these samples is set aside as a calibration set, labelled $((X_1, Y_1), \dots, (X_m, Y_m))$. A black-box segmentation model $f : \mathcal{X} \rightarrow [0, 1]^{d \times d \times d \times k}$ is then trained to produce an initial mask $f(X_i)$, where $f(X_i)_y$ is a k -dimensional confidence vector. These confidence scores take the form of per-voxel pseudo-probabilities summing to 1 across the possible classes (e.g., background/no tumour vs. tumour subtypes). The conformal layer then interprets these confidence scores and seeks to adjust the segmentation in a way that ensures the chosen risk measure, here FNR, remains below the specified level α in expectation. To form a conformal set, the classes are sorted by their corresponding probabilities for each voxel in descending order. Denoting the sorted classes as $\pi_1(X), \pi_2(X), \dots, \pi_k(X)$, one can interpret $\pi_1(X)$ as the class with the highest model-assigned probability, $\pi_2(X)$ as the second highest, and so on.

A threshold λ is chosen based on the empirical quantiles of the calibration set. Formally, define:

$$C_\lambda(X_i)_y = \left\{ \pi_1(X_i), \dots, \pi_k(X_i) \text{ with } k \text{ s.t. : } \sum_{j=1}^k f(X_i)_{\pi_j(X_i)} \geq 1 - \lambda \right\}, \text{ for each voxel } y.$$

The summation inside this set notation is used to decide how many of the top classes are included in the conformal set based on their cumulative probability and the chosen threshold. The net effect is that for a higher confidence threshold $\hat{\lambda}$, fewer classes are selected, leading to more specific predictions. Conversely, a lower $\hat{\lambda}$ results in a larger or more inclusive set, decreasing the chance of missing tumour voxels but increasing the chance of labelling healthy tissue as tumour.

This conformal set construction is repeated for each voxel in every scan, resulting in an entire segmentation mask that also carries a measure of confidence: pixels where the set is small are places where the model is highly certain, whereas larger sets indicate more uncertainty. These segmentations are then evaluated to determine how well they meet the FNR criterion on the calibration set to confirm that, in expectation, it does not exceed α .

To formalise the approach, the false negative rate is defined as:

$$L_i^{\text{FNR}}(\lambda) = 1 - \frac{|Y_i \cap C_\lambda(X_{n+1})|}{|Y_i|},$$

where C_λ is the conformal set determined by a threshold λ . The mask $C_\lambda(X_{n+1})$ includes all voxels the model flags as tumour when the confidence score exceeds a certain boundary that depends on λ . The fraction

$$\frac{|Y_i \cap C_\lambda(X_{n+1})|}{|Y_i|}$$

represents the proportion of true tumour voxels that were correctly identified by the conformal set. Consequently, $L_i^{\text{FNR}}(\lambda)$ is the proportion of tumour voxels that were incorrectly labelled as non-tumour. A crucial feature here is that this loss function is bounded between 0 and 1 and is monotonic

with respect to the size of the segmentation mask C_λ : as λ becomes more permissive, fewer voxels are labeled non-tumour, reducing FNR but also potentially introducing more false positives.

Following the Conformal Risk Control framework, one can select $\hat{\lambda}$ from the calibration data such that:

$$\mathbb{E} \left[L_{n+1}^{\text{FNR}}(\hat{\lambda}) \right] \leq \alpha,$$

where the expectation is taken over new samples (X_{n+1}, Y_{n+1}) . Intuitively, the calibration procedure collects confidence scores on the known data, analyses their distribution, and determines a threshold $\hat{\lambda}$ that yields the empirical bound on FNR. Under mild assumptions that the calibration samples and new samples are exchangeable, we obtain a formal finite-sample guarantee on the expected false negative rate.

Model Architecture and Training

The base model used for this case study is adapted from Rastislav [5], which uses a U-Net architecture for segmentation of tumours in MRI images. Although the exact details of the architecture—such as network depth, choice of convolutional blocks, skip connections, and the use of techniques like attention or multi-scale feature extraction—are beyond the primary scope of this paper, it is important to note that our conformal prediction framework is largely model-agnostic. Any segmentation network capable of generating a confidence map for each voxel can be extended using our approach. In short, the baseline model might be a U-Net or a variant thereof, trained with a standard cross-entropy or Dice loss to distinguish among multiple tumour classes (e.g., healthy tissue, tumour type A, tumour type B).

For the case study, the publicly available BraTS2020 dataset [2, 3, 4] will be used, which comprises 368 multi-modal MRI scans with corresponding expert-annotated tumour segmentation masks. Following the original protocol, the dataset is divided into multiple subsets. First, 80% (294 scans) is allocated for model training, while the remaining 20% (74 scans) is reserved exclusively for final validation. The training-validation portion is further split, with 50% (147 scans) dedicated to training the neural network and the other 50% (147 scans) used to calibrate the conformal layer. Finally, the model training set is split again, allocating 85% (125 scans) for actual training and 15% (22 scans) for internal testing to evaluate generalisation. This structured partitioning ensures that calibration and validation data that remain independent of model training and selection which is crucial to evaluate the conformal method.

Interpretation of Uncertainty Regions

The advantage of this voxel-level conformal approach is that it provides a natural “uncertainty map” across the entire scan. Clinicians, radiologists, and automated systems can interpret this map to identify areas warranting further attention. For instance, if a localised area in the scan is assigned a set of classes that includes both “no tumour” and a rare tumour type, the framework highlights that the model is uncertain here, and a radiologist should probably investigate it more thoroughly. In effect, this approach merges the best of both worlds: a precise segmentation in regions where confidence is high and a modest but justifiable expansion in ambiguous areas to reduce the risk of excluding real tumour tissue.

2 Results

The experiments, using 368 MRI scans, evaluated the performance of the proposed conformal risk control approach in terms of FNR. Specifically, the hyper-parameter was set $\alpha = 0.05$, for an expected FNR of 5%. After training the baseline segmentation model, we used the 40% calibration split to derive our conformal threshold $\hat{\lambda}$. The calibration procedure examines the distribution of confidence scores and identifies $\hat{\lambda}$ that ensures at most 5% of tumour voxels are missed in expectation.

When the resulting threshold parameter $\hat{\lambda}$ is applied to the held-out MRI scans, the overall distribution of observed FNR values was centred around the 5% target, with only a small proportion of outliers

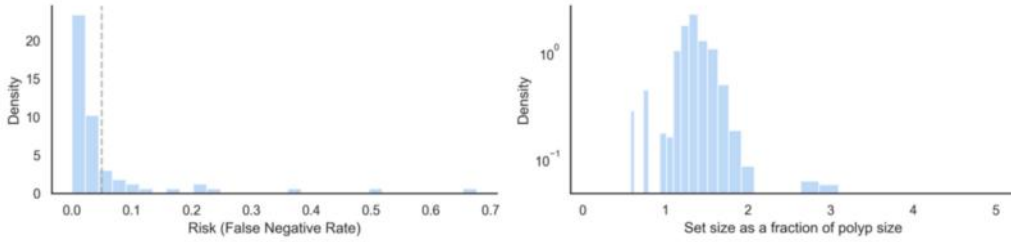


Figure 2: Panel A: Risk (FNR) distribution; Panel B: Histogram of conformal prediction set size in comparison to true tumour size.

exceeding the threshold. Figure 2 illustrates these outliers in the distribution of FNR, showing that most scans indeed have low FNR, while a minority still surpasses the 5% bound. Despite these outliers, the average FNR across the dataset was 5.58%, with a standard deviation of approximately 10.95%. This slightly higher value than the 5% target is partly due to finite-sample correction. Nonetheless, these results demonstrate that conformal risk control is capable of bounding the proportion of undetected tumour voxels, thereby addressing one of the most critical pitfalls in segmentation tasks.

Another important part was understanding how the conformal approach adjusts the size of the segmentation mask in order to control the false negatives. Figure 3 illustrates the relationship between the expanded segmentation masks and the ground-truth tumour masks across the full dataset. Typically, the conformal uncertainty sets slightly exceed the true tumour mask, introducing a layer of safety margin. This is an expected consequence of bounding the false negative rate.

While a slight inflation of the predicted tumour region is often acceptable in clinical practice (since false positives can be evaluated by a radiologist or further diagnostics), there is a practical limit to how large these expansions can become before they become unwieldy. Striking a balance between an acceptable false negative rate and a manageable number of false positives is thus part of the design space that clinicians can navigate by adjusting α or other hyper parameters in the conformal process.

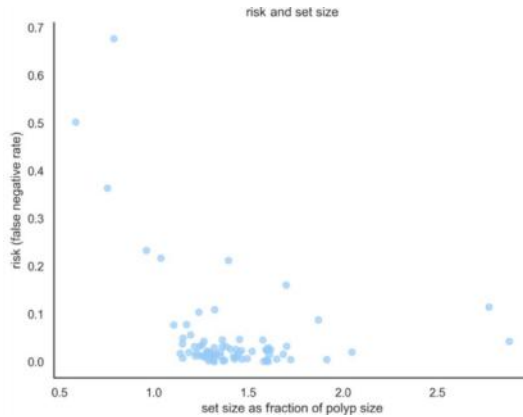


Figure 3: Spatial Distribution of FNR and set size as fraction of tumour size.

We visualized per-voxel errors in several representative MRI scans to identify patterns in how the conformal sets compensate for baseline segmentation mistakes. Figure 3 shows that scans with the most egregious FNR outliers typically also had a low overall set size. In other words, the places where the method failed were predominantly those where the baseline model had unwarranted high confidence in labelling a region as non-tumourous. Because the conformal expansion is triggered by model uncertainty, a miscalibrated or overly confident model may produce minimal expansions precisely where they are most necessary. This underscores the “garbage in, garbage out” principle: conformal prediction cannot correct large mistakes in the underlying model’s confidence distribution. However, in cases where the baseline model was moderately uncertain, the conformal mechanism effectively expanded the segmentation boundary, thus capturing otherwise missed tumour voxels.

Qualitative Visualisation

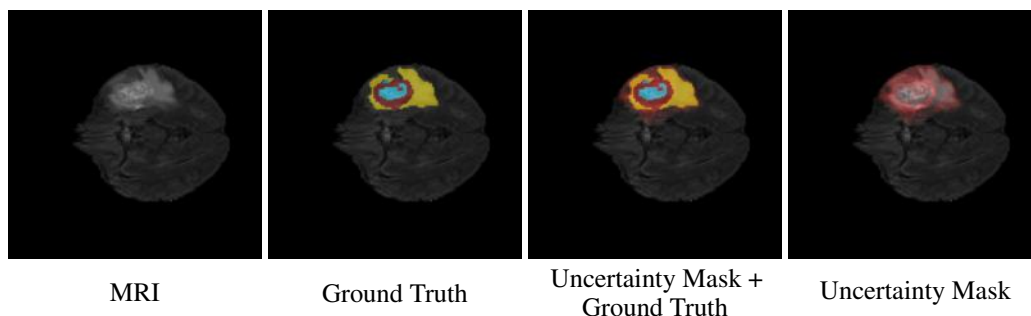


Figure 4: **A well-detected glioma.** (Case 93, Z-slice 59). The baseline model confidently and correctly detected and segmented the glioma, with small expansion of the uncertainty set along the boundary. Here, the conformal sets successfully mitigated borderline misclassification, assuring a low FNR without excessive false positives.

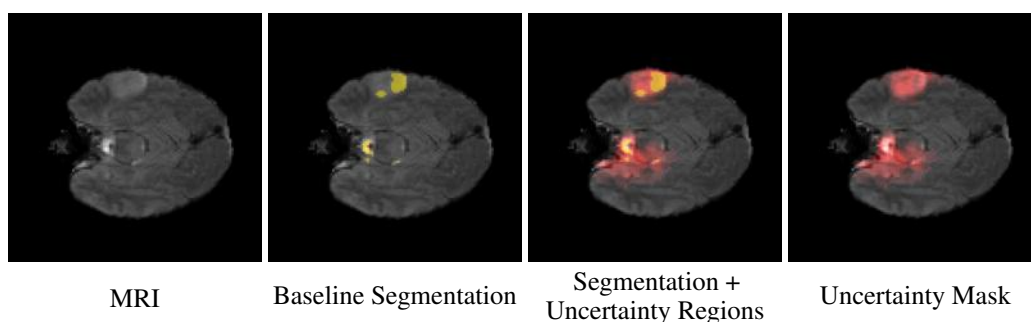


Figure 5: **A partially undetected tumour region corrected by uncertainty map.** (Case 70, Z-slice 25). The baseline segmentation (yellow) only partially detects the glioma (right temporal lobe). By integrating uncertainty quantification (red), the model explicitly identifies areas where the segmentation is unreliable, directing clinical attention to regions at high risk of being misclassified as non-tumorous.

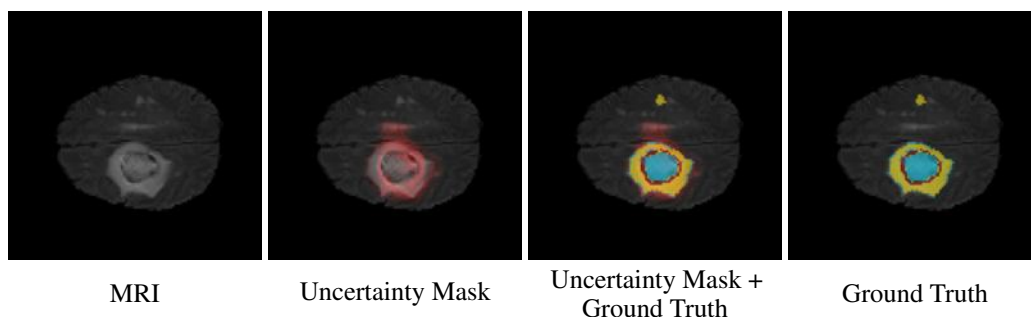


Figure 6: **An undetected tumour nodule misclassified with high confidence.** (Case 25, Z-slice 97). The baseline model missed a small secondary tumour nodule in the right cerebral hemisphere, classifying it 'non-tumorous' with high confidence. The conformal uncertainty mask is unable to adapt, resulting in an undetected tumour. This example highlights the limitation of relying on the baseline models confidence scores: if they are confidently incorrect, the conformal sets offer limited corrective ability.

Discussion

The empirical findings demonstrate that the proposed conformal framework can, on average, control the false negative rate at the desired threshold. This is a substantial improvement over conventional segmentation models that typically lack a mechanism to offer formal risk control. Many deep learning models offer probabilistic outputs which are often interpreted as “confidence”, but are merely pseudo-probabilities that do not provide robust uncertainty quantification to make claims on clinically significant metrics like the false negative rate. By providing a rigorous way to calibrate these models, it is possible to equip clinicians with quantifiable risk estimates. The interpretability of voxel-level confidence expansions can reduce the likelihood of missing tumour regions, which is a primary concern in clinical scenarios.

However, there are a few key considerations. The approach heavily relies on the quality of the baseline model’s ‘confidence’ scores. Conformal prediction can only expand or contract the segmentation boundaries based on the internal uncertainty distribution provided by the black-box model. If the model is systematically overconfident in misclassified voxels, conformal prediction can not rectify the errors. This was clearly observed in the analysis of outliers with high FNR. The “garbage in, garbage out” principle thus holds: strong performance relies on well-trained, properly calibrated networks.

In practice, controlling the FNR naturally increases the size of the uncertainty sets. Most medical imaging contexts can tolerate some degree of overestimation, since suspicious areas can be investigated through visual inspection. Nonetheless, an excessively inflated segmentation mask might be problematic, wasting time and resources. The threshold parameter α can be adjusted to find an appropriate trade-off. Lower α yields fewer false negatives but at the cost of a larger set size. Setting α too high diminishes the method’s utility by allowing more missed tumour voxels. Therefore, the final choice of α should be guided by the specific application context.

Although the case study presented here focuses on tumour segmentation, the methodology is easily applicable to other domains in medical imaging where robust uncertainty quantification can be beneficial. Examples include detecting internal bleeding in emergency radiology, identifying infection regions in lung CT scans, or segmenting pathological structures in neurological disorders. In fact, any task for which a (black-box) model and a monotonic loss function is used may benefit from conformal risk control.

It is also worth exploring the potential of integrating domain specific knowledge in the conformal framework. In some cases, prior knowledge about tumour growth patterns, typical shapes, or adjacency constraints could further refine the segmentation approach. Rather than purely relying on voxel-level confidence scores, advanced conformal strategies could incorporate spatial regularization terms, morphological constraints, or region-based confidence aggregation. Such additions may reduce both false positives and false negatives, improving the clinical utility of the method.

Finally, there remain open theoretical questions regarding conformal prediction in high-dimensional image data. While the present approach is grounded in well-established finite-sample guarantees, these results typically presume exchangeability and independence assumptions that may not fully hold in medical datasets exhibiting correlated structures across scans or within scans. Nonetheless, in many practical scenarios, the approximate exchangeability assumption remains a valuable tool, and the empirical performance results suggest that the method is robust enough for many real-world applications.

Conclusion

This paper introduced a rigorous way to incorporate conformal prediction into tumour segmentation systems, providing an additional layer of reliability and interpretability to black-box models. By focusing on the false negative rate as the primary risk metric, the paper has addressed one of the most serious concerns in medical imaging: failing to detect tumour regions that require clinical intervention. The approach utilises Conformal Risk Control to create segmentation masks augmented with uncertainty masks, ensuring that on average the fraction of missed tumour voxels remains below a user-specified threshold.

Through experiments on a dataset of 368 MRI scans, it is demonstrated that this framework can keep the average FNR around 5% with modest variability. The methodology is simple to implement and

computationally inexpensive, requiring only a reliable calibration set that is distinct from the data used to train the segmentation model. The flexibility of the approach in controlling different types of errors makes it valuable for a wide range of use cases in healthcare and medical imaging.

Nevertheless, certain limitations highlight the need for future improvements. In particular, miscalibrated or overconfident baseline models pose a significant challenge, as the conformal uncertainty masks do not inherently fix confidently incorrect errors. Furthermore, while bounding the FNR can be a lifesaving advantage in critical medical applications, this comes at the cost of extra false positives, that might draw away the attention of the clinician. The ongoing challenge will be to refine these methods and baseline models to create more robust uncertainty metrics for the conformal method and to further improve accuracy and confidence.

Overall, the results suggest that integrating conformal prediction into segmentation systems holds significant promise for enhanced medical decision-making. By quantifying uncertainty at each voxel while constraining the overall error rate, clinicians gain interpretable, controllable predictive models. As deep learning in healthcare continues to develop, robust uncertainty quantification will be essential to building trust, and to ultimately provide better patient outcomes. The framework presented here is one step toward making these models more transparent, reliable, and clinically relevant. It is therefore, anticipated that conformal prediction will see increasing adoption in high-stakes settings where risk control cannot be an afterthought.

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Appendix

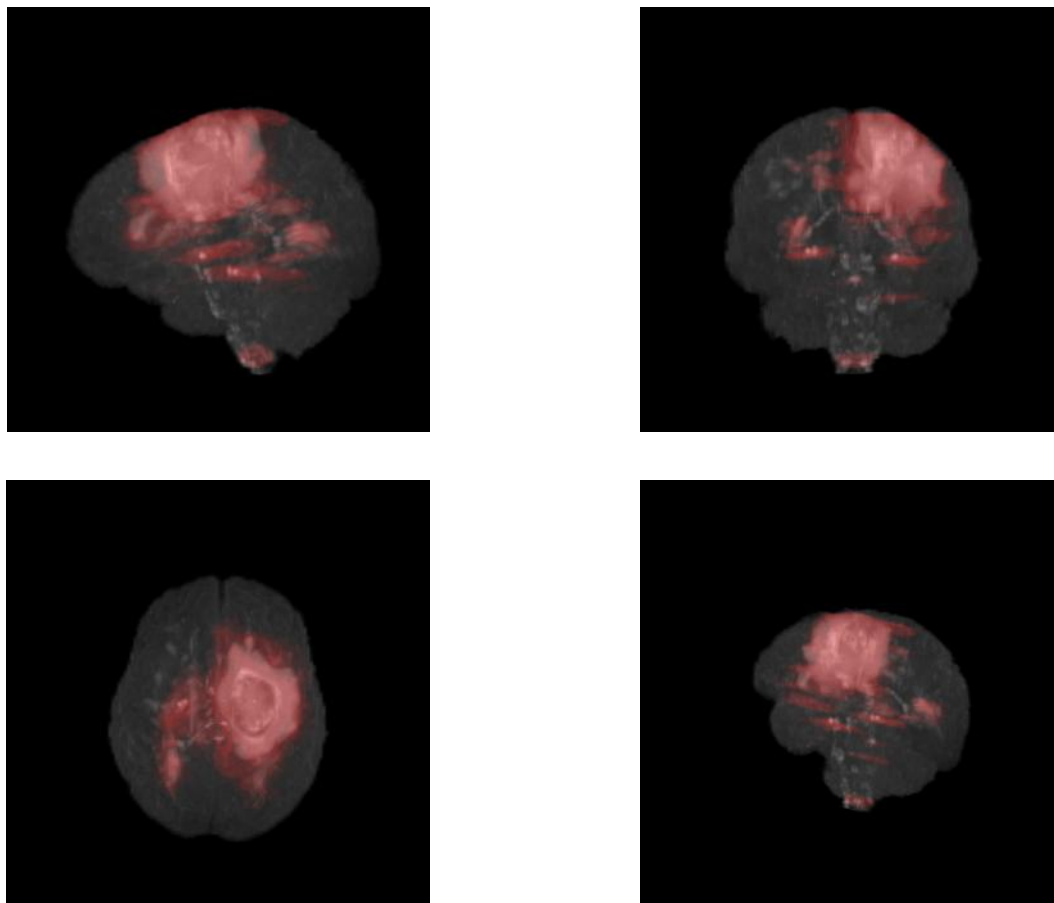


Figure 7: Visualisation of the 3D MRI imaging tool developed for this project, demonstrating an MRI scan (case 25) with an overlaid uncertainty mask to indicate regions of segmentation uncertainty.